

Rakesh Reddy Yeduru

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Summary

Data Engineer with 3+ years of hands-on experience translating business requirements into scalable, automated data solutions using PowerBI, Python, SQL, and Hadoop. Developed robust ETL pipelines and enhanced data warehouse capabilities in fast-paced environments. Proven track record in optimizing data processing, integrating advanced tools such as Apache Airflow, and employing Unix/Linux for efficient scripting and automation. Eager to contribute technical expertise to drive impactful data management initiatives.

Work Experience

Community Dreams Foundation, Florida, USA December 2024 - Present

Data Engineer

- Managed and audited large volumes of student housing data and internal databases, ensuring accuracy and compliance with policies.
- Automated room assignment, occupancy reporting, and waitlist tracking using Python and SQL to deliver technical solutions aligned with evolving business needs, improving operational efficiency by 25%.
- Designed and deployed ETL pipelines and interactive PowerBI dashboards with Apache Airflow for job scheduling and automation, enhancing data warehouse robustness and scalability.
- Integrated housing datasets into a centralized Snowflake data warehouse, streamlining data analysis while supporting detailed data modeling.
- Developed reusable SQL scripts and Python automation tools to reduce manual reporting tasks and support efficient production data workflows.
- Implemented version control via GitHub to ensure collaborative development, traceability, and adherence to CI/CD best practices.

Macy's, USA August 2023 - November 2024

Data Engineer

- Developed and maintained batch and real-time data processing systems using Hadoop, PySpark, and Kafka.
- Designed and implemented SSIS packages to integrate data from multiple sources including Excel, SQL Server, and flat files.
- Wrote advanced SQL scripts with secure credential handling via Azure Key Vault to query and manipulate large datasets.
- Developed Hive scripts and MapReduce programs to support data processing and analytics workloads.
- Collaborated with analytics teams to integrate processed data into Snowflake for BI reporting and analysis.
- Conducted tool evaluations (PyCharm, Jupyter Notebook) to enhance development productivity and code quality.
- Created Tableau dashboards to visualize key business metrics, improving decision-making speed and accuracy.

LTIMindtree, India July 2022 – July 2023

Data Engineer

- Designed and implemented ETL pipelines using PySpark and Apache Airflow to process structured and semi structured data into Snowflake.
- Migrated on-premises databases to Azure Data Lake, ensuring minimal downtime and high data integrity.
- Optimized SQL queries and implemented indexing strategies, reducing query execution times by 40%.
- Built Tableau dashboards for real-time analytics, delivering actionable insights to business teams.
- Developed and maintained incremental load processes to improve data pipeline efficiency and reduce resource consumption.
- Created automated data quality checks to detect anomalies before loading data into production systems.
- Integrated GitHub for CI/CD workflows, improving collaboration and version control in data engineering projects.

Indian Institute of Technology, India January 2022 - July 2022

Data Analyst Intern

- Developed an LSTM-based phishing URL detection model with 96% accuracy using PyTorch and advanced embeddings.
- Engineered adversarial defense techniques to improve CNN model resilience from 82% to 98% accuracy.
- Designed feature engineering workflows integrating Python and Spark for scalable ML preprocessing.
- Conducted data cleaning, normalization, and augmentation for deep learning datasets.
- Utilized GitHub for version control and collaborative development.

Cisco Networking Academy, India March 2021 - June 2021

Cyber Security Architect Intern

- Simulate university-wide network infrastructure using Cisco Packet Tracer, applying advanced topologies and layered security protocols.
- Establish secure network architectures and implement cybersecurity protocols across complex topologies. Conduct network security audits and risk assessments to identify and mitigate vulnerabilities in network configurations, applying resolution strategies.
- Collaborate in Agile teams to maintain a 98% network uptime by analyzing threats, detecting intrusions, and configuring secure TCP/IP.

Skills

Programming & Development: Python, SQL, MySQL, MongoDB, C++, Machine Learning Algorithms.

Big Data & Cloud: AWS, Azure Data Factory, Snowflake, PySpark.

Data Engineering & Analytics: Apache Spark, Kafka, Hadoop, Airflow, Data Migration, Tableau, Power BI.

Frameworks & IDE's: TensorFlow, PyTorch, Keras, Pandas, NumPy, OpenCV, Matplotlib, Terraform, Jenkins, CI/CD, Jira, VS Code, PyCharm.

ETL Tools & DevOps: SSIS, Informatica, Docker, Kubernetes.

Version Control: GitHub, Git, GitLab.

Education

University of Dayton Dayton, OH, USA
Master of Science in Computer Science GPA: 3.74 | August 2025

Projects

Azure Healthcare Data Lakehouse

- Integrated EMR, lab, and IoT device data into Azure Data Lake + Snowflake, with Hadoop + PySpark for large-scale processing.
- Orchestrated ETL via Airflow, containerized with Docker, and deployed to Kubernetes with Terraform automation.
- Delivered Power BI dashboards, reducing clinical reporting delays by 40%.

Supply Chain Optimization Dashboard Using PowerBI

- Unified ERP and vendor data from SQL Server + Snowflake, enabling advanced Power BI dashboards with DAX measures.
- Automated refresh pipelines with Airflow/Linux, containerized ETL via Docker/Kubernetes, and managed infra with Terraform.
- Improved efficiency, driving a 15% increase in on-time delivery and 12% lower logistics costs.

Real-Time Retail Analytics Warehouse Using AWS

- Developed a Kafka + AWS Kinesis pipeline with AWS Glue (PySpark) and Redshift, integrating a Hadoop batch layer for historical sales data.
- Containerized ETL with Docker/Kubernetes and automated deployments using Jenkins CI/CD.
- Built Tableau dashboards, reducing sales reporting latency from 24 hours to 5 minutes.